

MSc in High Performance Computing

MSc in High Performance Computing with Data Science

Programming Skills Coursework – Performance

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12 July 2016

1 Introduction

In this coursework you will carry out performance experiments using the code you, and your team, developed as part of Coursework - Development.

The coursework is an *individual* exercise. You will design and carry out these experiments yourself, and write up your experiments in a report.

Your mark is based upon your report.

2 Design and carry out performance experiments

Design and carry out suitable experiments to assess the performance of the code you produced as part of Coursework - Development. It is up to you what experiments you conduct, but some possibilities include:

- Investigate the effects of compiler optimisation flags.
- Identify and quantify the main sources of overhead.
- Investigate the effects of landscapes with different proportions of land and water.
- Investigate the performance for different sizes of landscape.

When investigating performance you should take care to justify your choice of metrics. You are not required to carry out all of the experiments above, nor to restrict yourself to this list. Quality is more important than quantity. Credit will be given for intelligent and insightful analysis of your results, in preference to the quantity of data you present. Graphical presentation of performance data is strongly encouraged.

3 Write your individual report

Your report should be structured as follows:

- **Introduction:** Short introduction.

- **Performance tests and analysis:** Description on the performance experiments you did and an analysis of the results of these.
- **Conclusions:** Some brief conclusions.

Your report should be about 4-6 pages.

4 Submission and marking

You are required to submit a written report.

Submissions are done via the Turnitin submission tool from within LEARN, under Coursework Submission.

The filename of your submission must include your *exam number* which is the B number which appears on your matriculation card (e.g. B123456). You should also include the course identifier, PS for Programming Skills, and the identifier PF for this coursework. For example B123456-PS-PF .pdf.

Your exam number should be used as the author of your report. Do not put your name on your report. This naming scheme helps to ensure that, wherever possible, we can mark coursework anonymously but can also easily match the final marks to the correct students.

4.1 Deadline

The deadline for submission for your report is **14:00 Friday 25th November (week 10) 2016** for all students.

Submission are allowed up to 7 calendar days after the deadline, with a 5% deduction per day or part-day late.

If you submit any coursework by the deadline then it will be marked, and you cannot submit an amended version later on, to be marked.

If you do not wish the submitted version to be marked after the deadline, and intend submitting an amended version after the deadline, then it is your responsibility to let the course organiser know, in advance of the deadline, that the submitted version should not be marked.

4.2 Marking

Marks will be awarded as follows:

- Description of performance analysis and results: 35
- Presentation (quality of text and report): 5
- **Total: 40**

4.3 Provisional marks

Your provisional marks and feedback will be returned no later than 14:00 Friday 16th December 2016.

5 Questions

Feel free to e-mail the course organiser if you have any questions. Questions of interest to the whole class will be anonymised and them, plus their answer, forwarded to the whole class.